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LISTS / COLLECTIONS

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What is a LIST?

A **LIST** is a container that can store **multiple values together in one place**.

Instead of creating many variables like:

SET MARK1 TO 80
SET MARK2 TO 70
SET MARK3 TO 90

You can store everything inside one LIST:

SET MARKS TO LIST [80, 70, 90]

Think of a LIST like:

🧺 A basket holding multiple items
📚 A row of books on a shelf
🗄️ A group of records in a file

LIST Terms

LIST → Collection of values
ADD → Insert a new value
REMOVE → Delete a value
SIZE → Total number of elements inside

✅ **Example 1 — Student Marks (Basic)**

SET MARKS TO LIST [80, 70, 90]

DISPLAY SIZE OF MARKS

Output:

3

Because there are 3 values in the list.

✅ **Example 2 — Adding Values**

```
SET MARKS TO LIST [80, 70]

ADD 90 TO MARKS

DISPLAY SIZE OF MARKS
```

Now list becomes:

[80, 70, 90]

✅ Example 3 — Removing Values

```
SET NUMBERS TO LIST [10, 20, 30]

REMOVE 20 FROM NUMBERS

DISPLAY SIZE OF NUMBERS
```

Now list becomes:

[10, 30]

✅ Example 4 — Using LIST with Loop

You can repeat through a list using SIZE.

```
SET MARKS TO LIST [50, 60, 70]

REPEAT FROM I = 1 TO SIZE OF MARKS
  DISPLAY MARKS[I]
END REPEAT
```

This displays all values one by one.

✅ Example 5 — Adding Input to List

```
SET NAMES TO LIST []

DISPLAY Enter Name
READ NAME

ADD NAME TO NAMES

DISPLAY SIZE OF NAMES
```

✅ Example 6 — Counting Passed Students

```
SET MARKS TO LIST [40, 55, 70, 30]

SET PASS_COUNT TO 0

REPEAT FROM I = 1 TO SIZE OF MARKS

  IF MARKS[I] GREATER THAN OR EQUAL TO 50 THEN
    SET PASS_COUNT TO PASS_COUNT + 1
  END IF

END REPEAT

DISPLAY PASS_COUNT
```

✅ Example 7 — Shopping List System

```
SET ITEMS TO LIST []

ADD Milk TO ITEMS
ADD Bread TO ITEMS
ADD Eggs TO ITEMS

DISPLAY SIZE OF ITEMS
```

🧠 When Should We Use LIST?

Use LIST when:

✓ You need to store many values
✓ Number of items may change
✓ You want to loop through data
✓ You want cleaner code

Instead of:

```
NAME1 NAME2 NAME3 NAME4 NAME5
```

You use:

```
SET NAMES TO LIST [...]
```

⭐ Real-World Examples

LIST can represent:

- Student marks
- Product inventory
- Bank transactions
- Parking slots
- Exam scores
- Employee names
- Messages
- Orders

Almost every real software uses collections.

🚀 Self-Practice Exercises

Students can try:

Exercise A

Create a list of 5 numbers and display size.

Exercise B

Add 3 names to a list and display them using loop.

Exercise C

Remove a number from list and show new size.

Exercise D (Logic)

Count how many numbers are greater than 50 in a list.